LungLogic: Intelligent Airflow Control for Realistic Breathing Simulation

Yuqiao LI
Chun Yin Ku (John)
Jiaquan Zhong

Presentation Overview

01

Mechanical Part

Pneumatic architecture
Airflow sizing
Physical prototype

02

Electrical Part

Power and bridge board
Sensor interfaces
Signal verification

03

Control Software & UI

Python interface
Reusable model + Auto PID
Hardware migration

Three parts, one system: pneumatic hardware, electrical interface, and reusable control software.

01

Mechanical Part

Pneumatic architecture • airflow sizing • physical prototype

Requirement

Objective: To develop a pneumatic control system and UI to drive a dynamic lung model, simulating controllable and repeatable respiratory motion.

12–16 bpm

Respiratory rate
nominal 15 bpm

500 mL

Nominal tidal volume

1:2

Nominal IE Ratio
1.3 s / 2.7 s

≥30 cycles

repeatability
variation < ±5%

Real-time UI

RR / VT / I:E
flow timing

Challenge

The lung model must still match the above requirement under the presence of back-pressure and airway resistance.

Team responsibilities & system architecture

01

Mechanical Part

Chun Yin Ku (John)

Pneumatic architecture · aerodynamic sizing
physical interface

02

Electrical Part

Jiaquan

Power distribution · PCB design
sensor interfacing

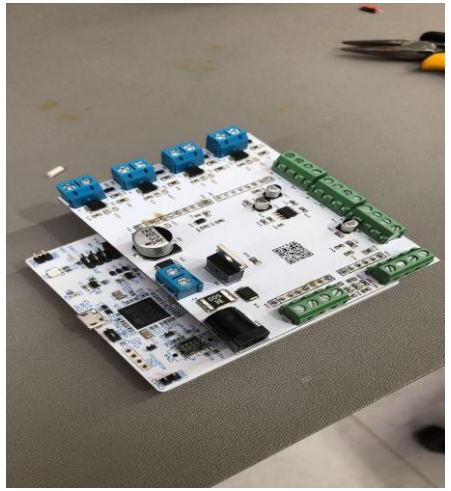
03

Control Software & UI

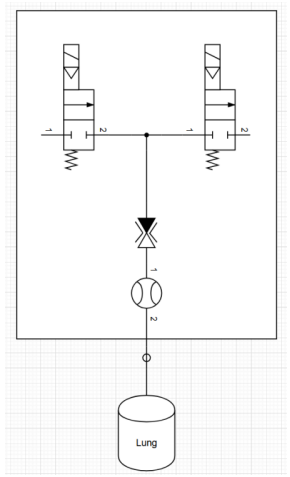
Yuqiao

STM32 control · flow-profile shaping
real-time UI

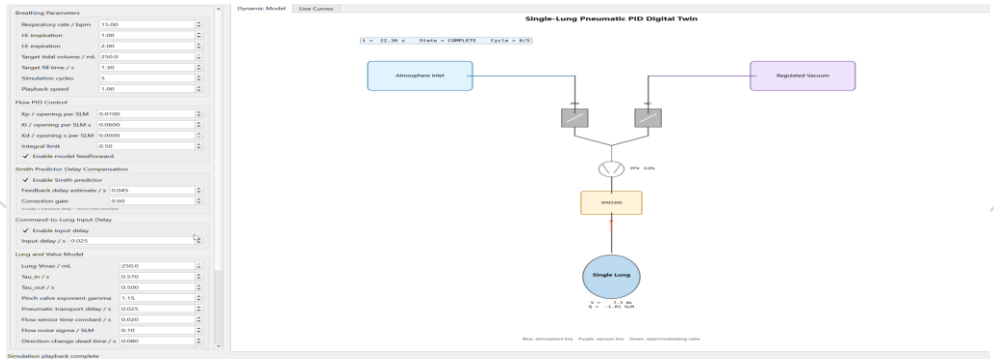
Integrated system



+

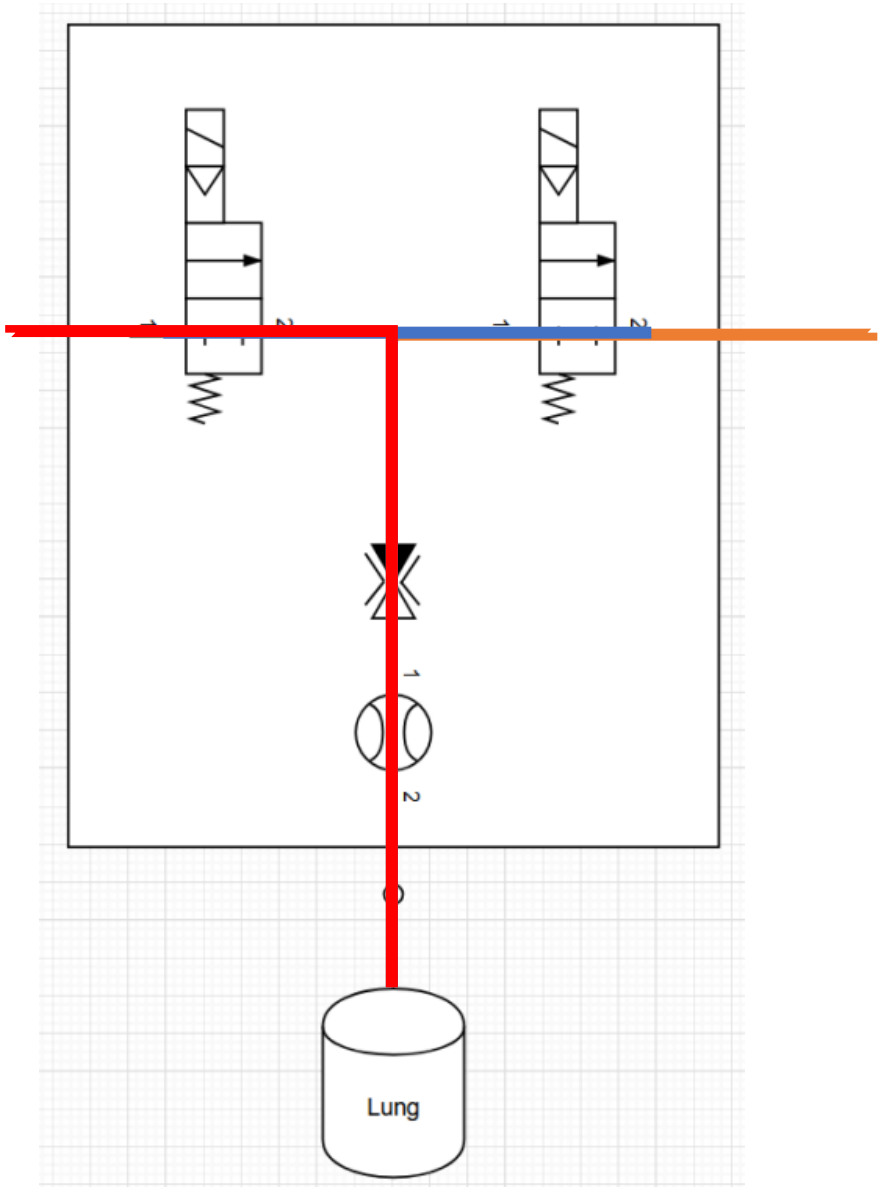


Control interface



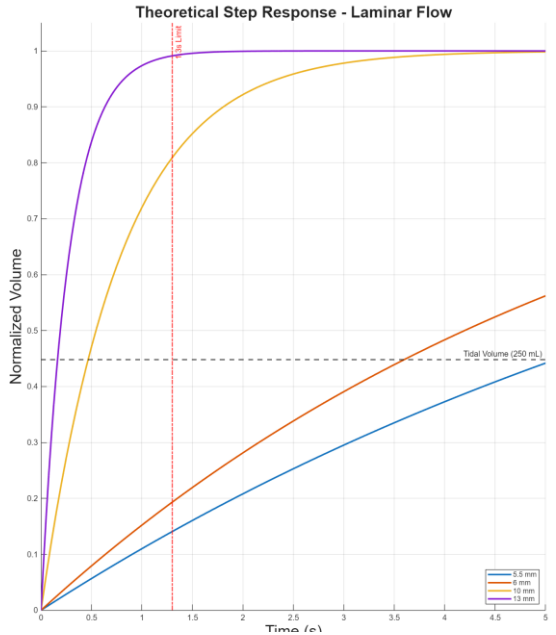
Objectives - Simulating Human Breathing

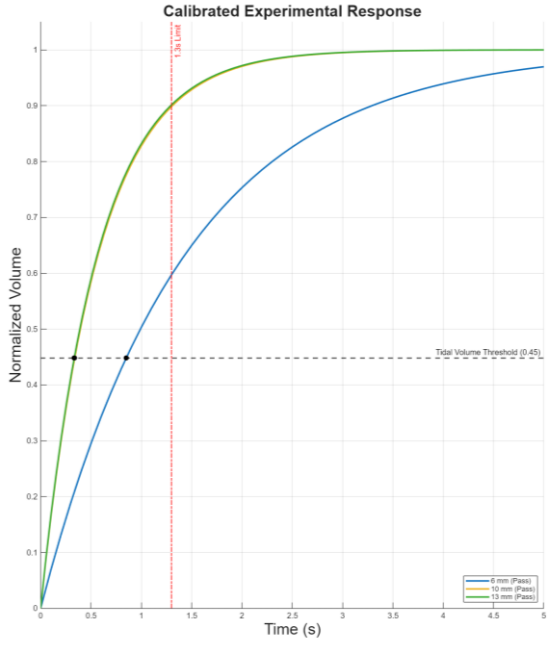
- **Goal:** Deliver 250mL volume per breath within realistic human timing ($\leq$ 1.3s)
- **Natural Mechanism:** Active vacuum simulates true diaphragm expansion, not positive-pressure blowing
- **Breath Phases:** Active expansion →natural hold →passive atmospheric release
- **Dynamic Control:** Precision constriction valve shapes smooth, realistic breath profiles



Current - Aerodynamic Optimization & Sizing Verification

- **Internal Diameter:** Standardized on 10 mm ID across the flow path to avoid choking
- **Filling Speed:** Delivers 250 mL in ~0.34 s unthrottled, well within the 1.3 s limit
- **Operating Pressure:** Set to -40kPa gauge operating vacuum
- **Tubing Buckling Limit:** Silicone collapses at -75kPa, providing a 1.9× safety factor

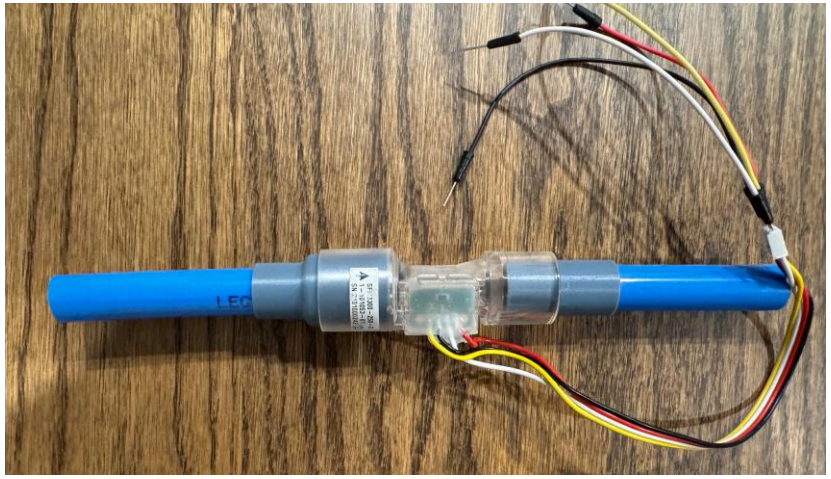




Current - Hardware & Lung Connection

- **Plug-and-Play Port:** Universal quick-connect coupling for the lung phantom
- **Breath Tracking:** Digital medical flow sensor continuously measures volume in real time
- **Custom Adapters:** Airtight 3D-printed housing connects sensors directly to the air line
- **Verification Phantom:** Physical test lung fabricated to validate airflow mechanics





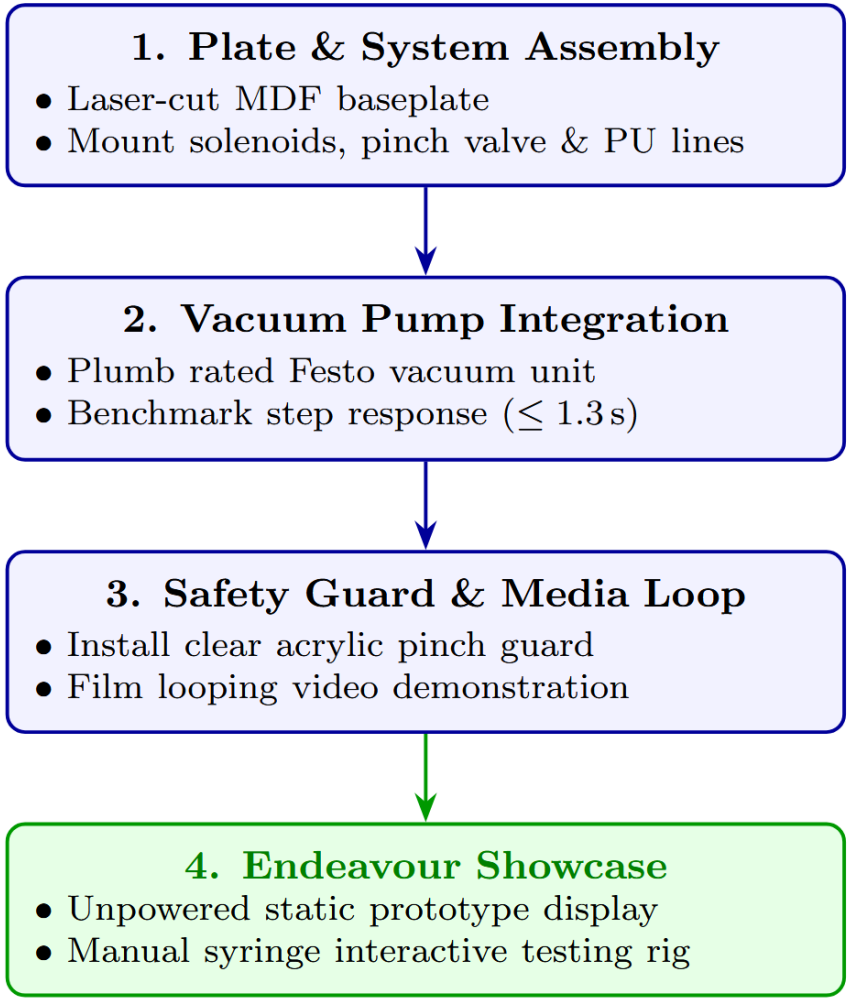
Current - Airtightness Quality Testing

- **Syringe Snap-Back Test:** Rapid, non-destructive vacuum check using a manual syringe
- **Airtight Benchmark:** 5-minute hold confirms zero atmospheric air leaks into the lines
- **Subsystem Isolation:** Test lung, custom adapters, and valves verified individually
- **Clean Process:** Avoids wet chemical sprays to protect sensitive sensors and electronics



Future - Final Assembly & Exhibition Plan

- **1. Plate & System Assembly:** Laser-cut MDF baseplate, mount solenoids, pinch valve & PU lines
- **2. Vacuum Pump Integration:** Plumb rated Festo vacuum unit, benchmark step response (≤ 1.3s)
- **3. Safety Guard & Media Loop:** Install clear acrylic pinch guard, film looping video demonstration
- **4. Endeavour Showcase:** Unpowered static prototype display, manual syringe interactive testing rig

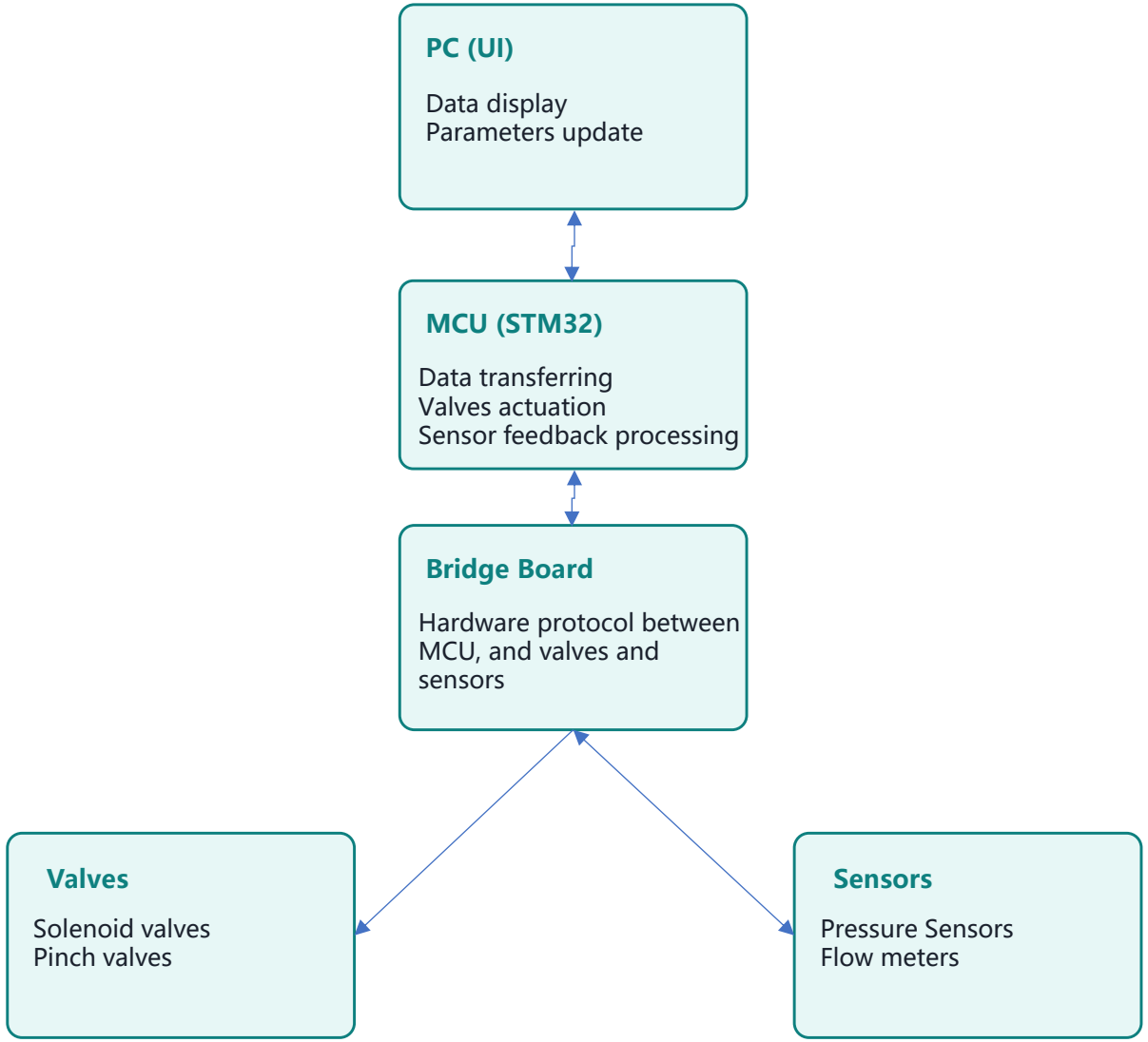


02

Electrical Part

Power distribution • bridge board • sensor and valve signals

Hardware Control Architecture



Bridging Board:

- **24V Input:**

Equipped with reverse polarity protection and overvoltage protection

- **Sensor Interface:**

I2C, 2 analogue channels with 180Hz LP filter

- **Emergency Response:**

Emergency button and feedback

- **4 Solenoid Drivers**

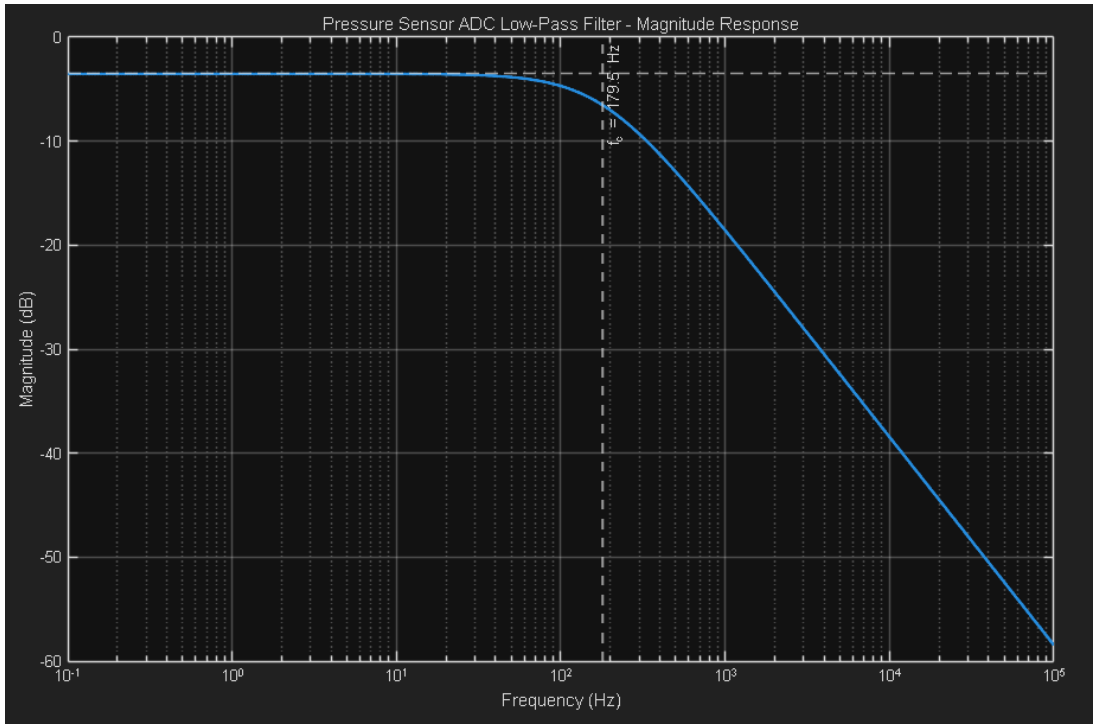
NMOS based GPIO driven

- **2 Pinch Drivers**

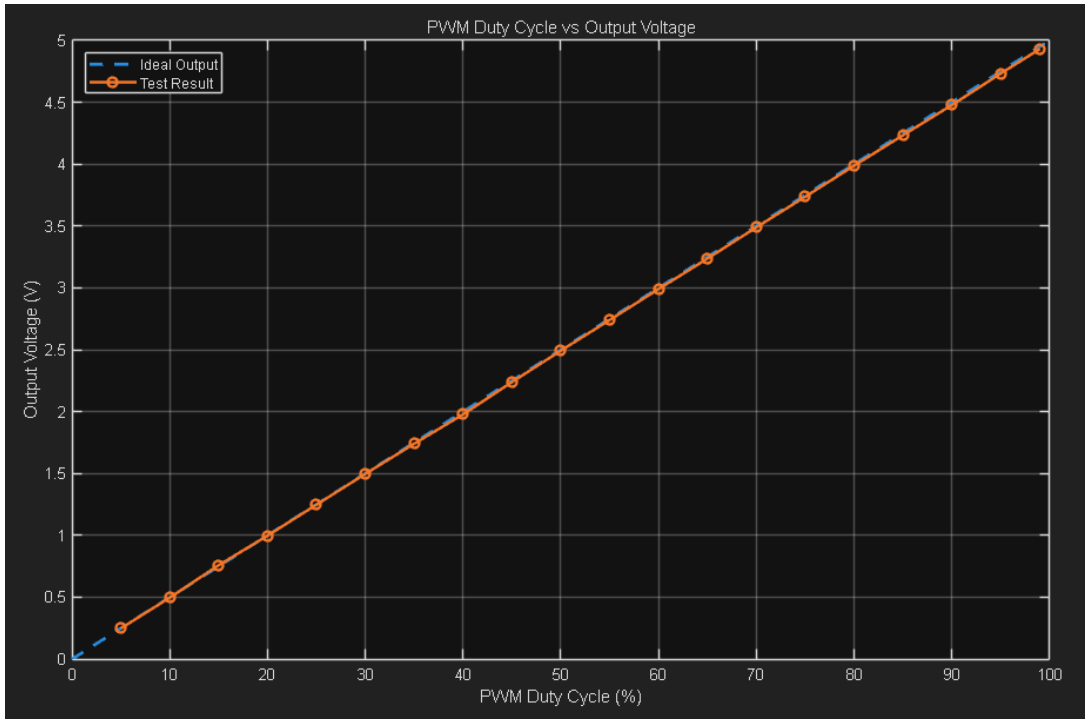
PWM (0-3.3V) driven with non-inverting proportional amplifiers

Signal verification:

180Hz Low Pass filter



PWM driven pinch control signal (0-5V)



03

Control Software & UI

Python interface • adjustable model • Auto PID • reusable workflow

Objective & requirements: a visible, adjustable Python control tool

OBJECTIVE

Visualise each breath and let users adjust the operating target without changing control code.

Breathing

Rate · tidal volume
I:E ratio · flow timing

Repeatability

≥30 cycles
within the agreed tolerance

Two interface levels

User Mode for operation
Developer Mode for tuning

PY

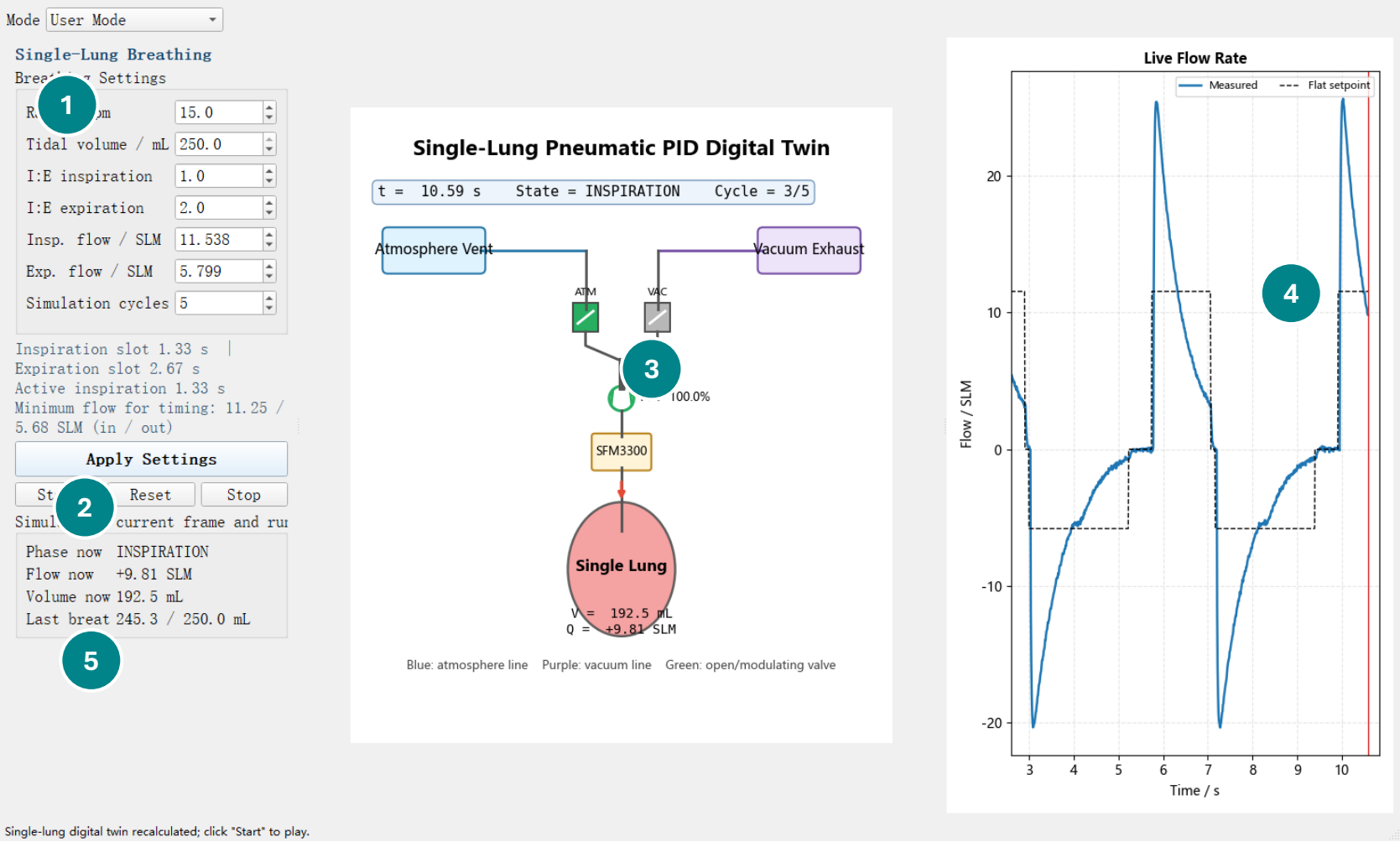
Python

WHY PYTHON

Fast UI iteration • scientific plotting • configurable models • repeatable tests • future hardware communication

Prototype scope today: one 250 mL lung branch; the software structure is designed to scale and be reused.

User Mode: configure and review a breathing cycle



1 Breathing settings

Rate, volume, I:E, flow and cycle count.

2 Apply & playback

Recalculate, start, pause, reset or stop.

3 Pneumatic state

Route, valve opening, phase, flow and volume.

4 Live flow

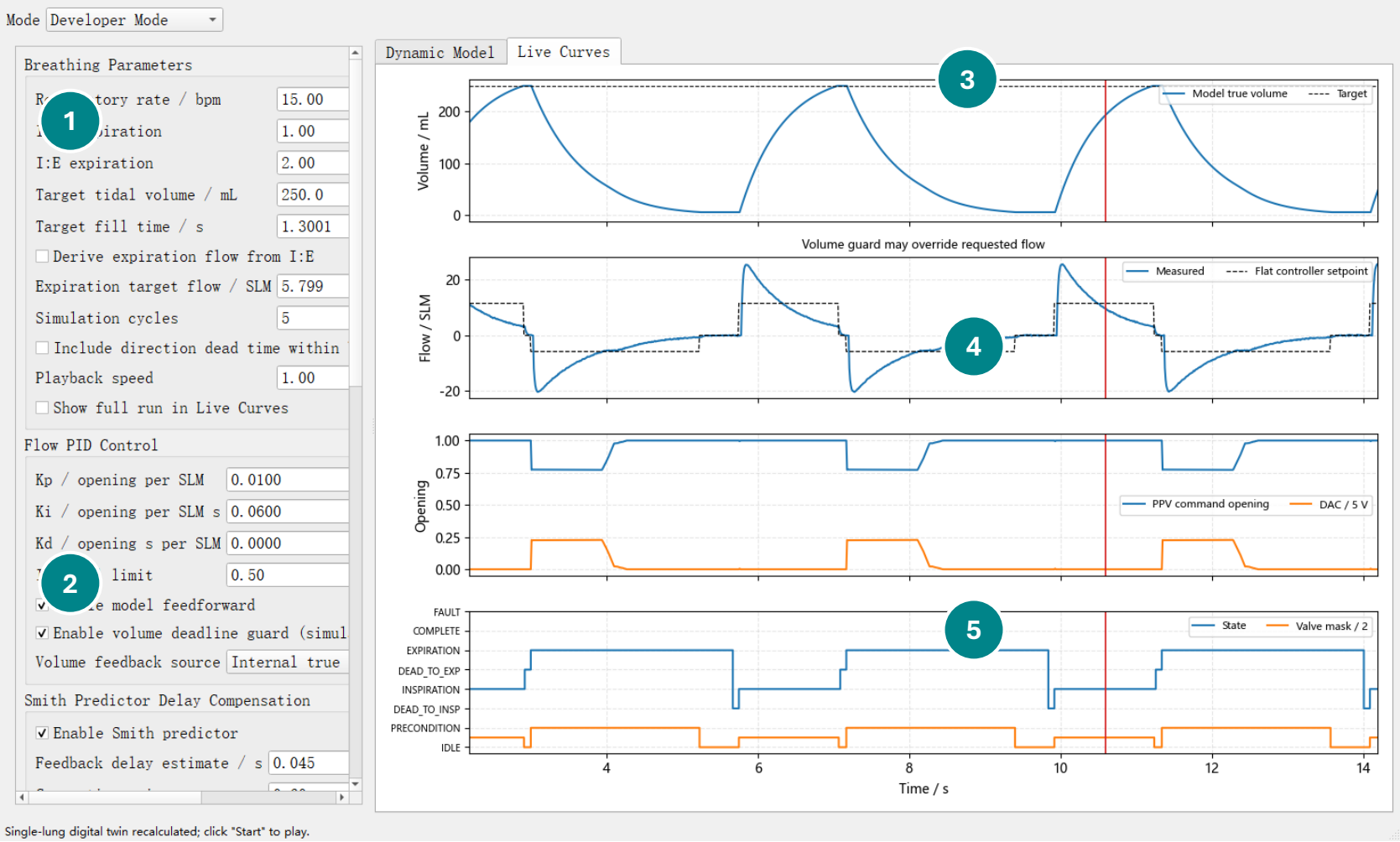
Simulated sensor flow versus requested setpoint.

5 Result

Current frame and achieved breath versus target.

User Mode keeps operation simple: choose the breath, apply it, and review the result.

Developer Mode: tune, diagnose and calibrate



1 Test conditions

Timing, target volume and number of cycles.

2 Control settings

PID, feedforward, Smith predictor and guard.

3 Volume trace

Target and repeated volume response.

4 Flow trace

Tracking, overshoot and response shape.

5 Actuation & state

PPV command, DAC output, route and timing.

Developer Mode links performance to the controller, model assumptions, delays and valve commands.

Control model: simple enough to tune, detailed enough to transfer

Breathing target
RR · VT · I:E
flow timing

→

State machine
ATM / VAC routing
direction dead time

→

PID + model tools
PI/PID · feedforward
Smith prediction

→

Lung + sensor model
capacity · time constants
delay · valve response

Q(t)
V(t)
commands

Adjustable model parameters

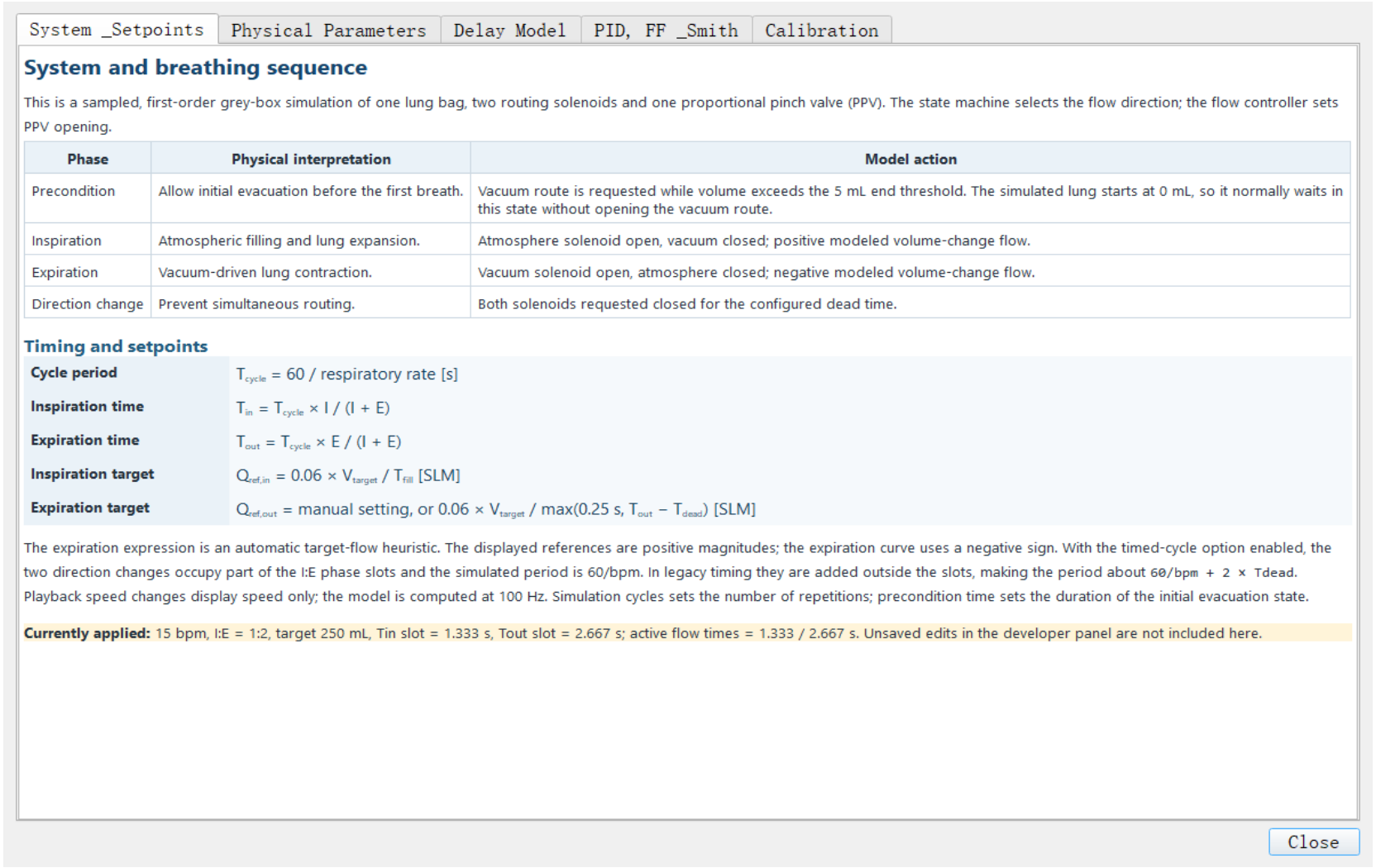
- Lung capacity and inspiration/expiration response
- Pinch-valve nonlinearity and command delay
- Sensor delay, response time, noise and zero bias
- PID gains, feedforward and prediction settings

AUTO PID — Offline candidate search

1. Tests Kp / Ki candidates
2. Scores 30-cycle results
3. Checks error, consistency and saturation
4. User reviews the best gains

Auto PID supports tuning; physical measurements still define the model and safe operating limits.

Built-in modeling & control guide



System _Setpoints | Physical Parameters | Delay Model | PID, FF _Smith | Calibration

System and breathing sequence

This is a sampled, first-order grey-box simulation of one lung bag, two routing solenoids and one proportional pinch valve (PPV). The state machine selects the flow direction; the flow controller sets PPV opening.

Phase	Physical interpretation	Model action
Precondition	Allow initial evacuation before the first breath.	Vacuum route is requested while volume exceeds the 5 mL end threshold. The simulated lung starts at 0 mL, so it normally waits in this state without opening the vacuum route.
Inspiration	Atmospheric filling and lung expansion.	Atmosphere solenoid open, vacuum closed; positive modeled volume-change flow.
Expiration	Vacuum-driven lung contraction.	Vacuum solenoid open, atmosphere closed; negative modeled volume-change flow.
Direction change	Prevent simultaneous routing.	Both solenoids requested closed for the configured dead time.

Timing and setpoints

Cycle period	$T_{cycle} = 60 / \text{respiratory rate [s]}$
Inspiration time	$T_{in} = T_{cycle} \times I / (I + E)$
Expiration time	$T_{out} = T_{cycle} \times E / (I + E)$
Inspiration target	$Q_{ref,in} = 0.06 \times V_{target} / T_{fill}$ [SLM]
Expiration target	$Q_{ref,out}$ = manual setting, or $0.06 \times V_{target} / \max(0.25\ s, T_{out} - T_{dead})$ [SLM]

The expiration expression is an automatic target-flow heuristic. The displayed references are positive magnitudes; the expiration curve uses a negative sign. With the timed-cycle option enabled, the two direction changes occupy part of the I:E phase slots and the simulated period is 60/bpm. In legacy timing they are added outside the slots, making the period about `60/bpm + 2 × Tdead`. Playback speed changes display speed only; the model is computed at 100 Hz. Simulation cycles sets the number of repetitions; precondition time sets the duration of the initial evacuation state.

Currently applied: 15 bpm, I:E = 1:2, target 250 mL, Tin slot = 1.333 s, Tout slot = 2.667 s; active flow times = 1.333 / 2.667 s. Unsaved edits in the developer panel are not included here.

Close

The guide explains

1. **System & setpoints**
   Breathing sequence and timing equations
2. **Physical parameters**
   Meaning and effect of every model input
3. **Delay model**
   Command, transport and sensor response
4. **PID, FF & Smith**
   How the control options interact
5. **Calibration**
   Which tests identify each parameter

The manual connects each adjustable value to its physical meaning, equation and calibration method.

Design philosophy: calibrate once, reuse the same framework

1

Connect a prototype

Different lung, tubing,
valves or sensor

→

2

Characterise it

Full-open response
Fixed PPV sweep

→

3

Update parameters

Capacity · τin · τout
γ · delays · limits

→

4

Validate & deploy

30-cycle check
Apply reviewed gains

REUSED

UI workflow · state machine · data logging validation sequence · documentation structure

RE-IDENTIFIED

Plant parameters · safe ranges · delays controller gains and hardware mappings

The prototype changes; the workflow stays. Measurements update the parameter file instead of rewriting the UI.

15 bpm achieved — now define the adjustable range

15.00 bpm

Completed validation

30 cycles	repeated simulation
242.6–246.3 mL	tidal-volume range
Within ±5%	of the 250 mL branch target

The nominal target is demonstrated in the current model benchmark.

INPUT NEEDED FROM 4DMEDICAL

What adjustable respiratory-rate range should the final system support?

1. **Minimum and maximum rate: 12–16, 12–18, or another range?**
2. **Required adjustment step and accuracy?**
3. **Must every rate hold the same 250 mL target and I:E = 1:2?**

Next requirement decision: define the operating envelope around the achieved 15 bpm nominal point.

Future work: migrate the model to the physical system

1 Calibrate the path

Full-open ATM/VAC response
Fixed-opening PPV sweep
Synchronize Q(t), V(t), P(t), commands

Measured model

→

2 Close the real loop

Drive PPV and routing valves
Feed SFM3300 data to controller
Verify timing and safety interlocks

Hardware closed loop

→

3 Validate performance

30 cycles at agreed nominal/boundary
Report volume, flow and saturation
Then test independent dual branches

Requirement evidence

Success is a measured 30-cycle result against the operating range agreed with 4DMedical.